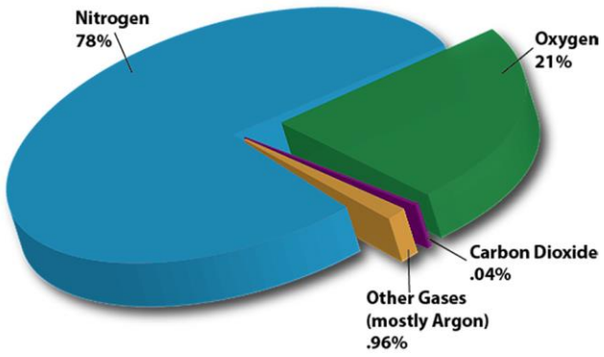
# Knowledge Organiser
# Year 7: Energy – Earth's Changing Atmosphere

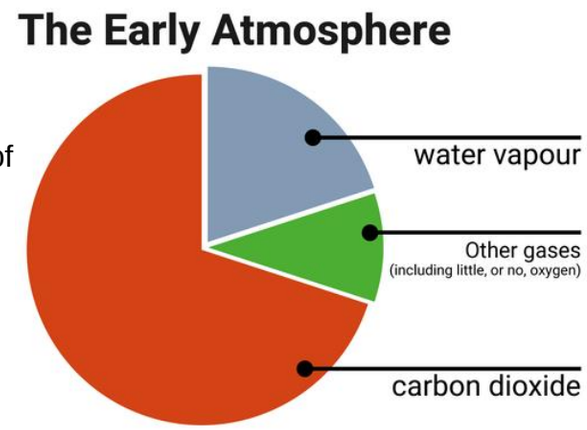
**What is Earth's atmosphere made of?**
- An atmosphere is a layer of gases which surround a planet
- Earth's atmosphere is a mixture of mostly Nitrogen and Oxygen gas
- Oxygen is needed for biological processes (like respiration), Nitrogen is an inert (unreactive) gas

A pie chart showing the approximate composition of the Earth's current atmosphere:




**How has the Earth's atmosphere changed over time?**
- The composition of the Earth's atmosphere hasn't always been the same
- In the past, the Earth's atmosphere was much more similar to the atmosphere on Mars today
- It was mostly carbon dioxide and water vapour (steam)

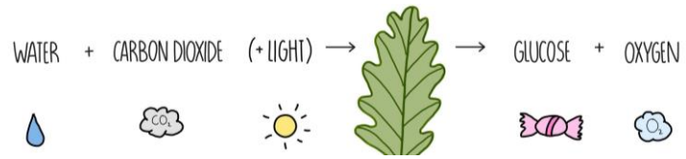
A pie chart showing the approximate composition of the Earth's earliest atmosphere:




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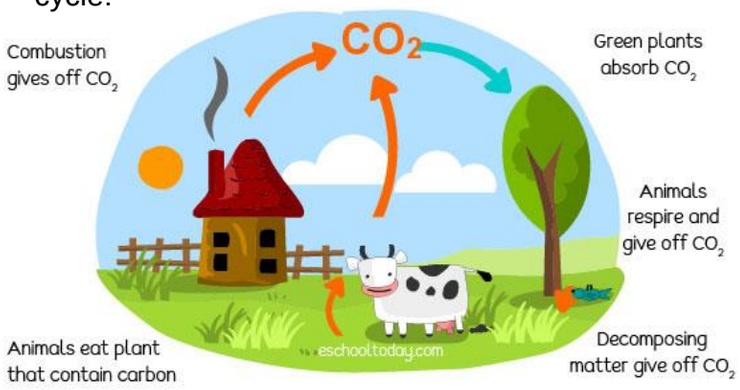
**What processes caused these changes?**
- As the Earth cooled over time, the water vapour in the air cooled and condensed
- This water formed the earliest oceans and lakes
- Eventually organisms evolved that used photosynthesis
- Photosynthesis uses carbon dioxide in the air to make stored sugars
- Volcanos continued to erupt, releasing large amounts of unreactive nitrogen gas

A chemical equation showing how photosynthesis removes carbon dioxide from the atmosphere




**What is the carbon cycle?**
- Carbon is an essential element for life, every living organisms has carbon compounds inside
- The carbon cycle is a series of processes that keep the levels of carbon dioxide in the atmosphere roughly the same
- Some processes like forest fires and volcanoes release carbon dioxide
- Others like photosynthesis and rock formation trap carbon dioxide

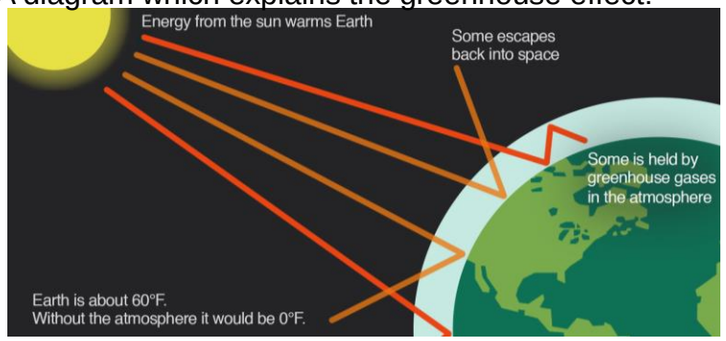
A diagram showing the main processes in the carbon cycle:




**How do human actions affect the carbon cycle?**
- Humans engage in behaviours which **release more** carbon dioxide than they trap
- Burning fossil fuels release $CO_2$ and the fuels will take millions of years to reform
- This means that levels of carbon dioxide in the atmosphere are steadily rising

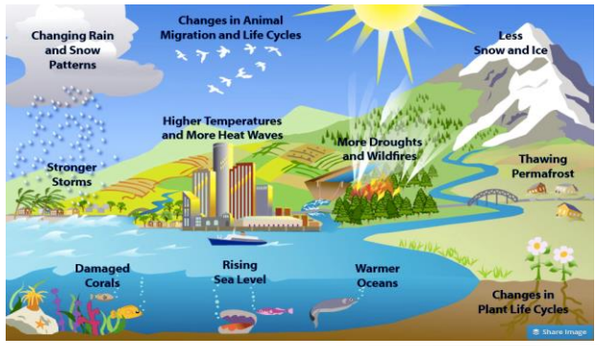
**How about the greenhouse effect?**
- A greenhouse works by trapping heat inside and keeping the plants warm
- The Earth's atmosphere does the same thing, trapping heat from the Sun
- Without the greenhouse effect the Earth would be too cold to live on

A diagram which explains the greenhouse effect:




**How does this cause global warming?**
- By increasing levels of $CO_2$, humans are increasing the levels of heat that get trapped
- This is causing **global warming** which is expected to increase temperature by at least 1.5 degrees by 2050

A diagram showing some of the impacts of global warming




## RAG Yourself!

| Learning Outcomes; | RAG |
|---|---|
| Draw a diagram showing the carbon cycle | |
| Describe how humans affect the carbon cycle | |
| Describe how $CO_2$ causes a greenhouse effect | |
| List some impacts of global warming | |
| State the composition of the atmosphere | |
| Describe the processes which have changed the Earth's atmosphere over time | |

# Knowledge Organiser
# Year 7: Energy – Earth's Changing Atmosphere

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A pie chart showing the approximate composition of the Earth's earliest atmosphere:

**What processes caused these changes?**

A chemical equation showing how photosynthesis removes carbon dioxide from the atmosphere

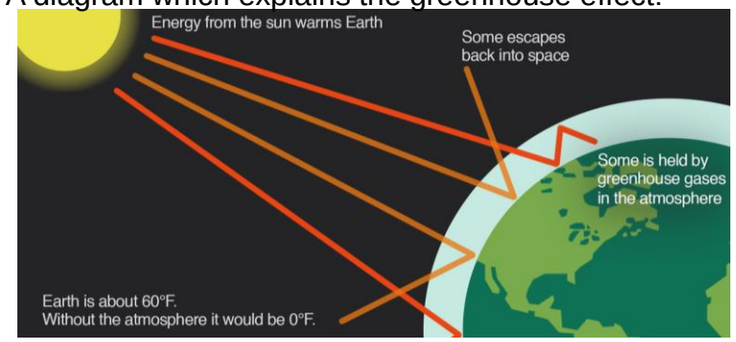
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A diagram which explains the greenhouse effect:




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